SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I J Praveen (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Problem Statement :

There are two unmarked jugs: one holds 4 gallons and the other holds 3 gallons. Using only the pump, the ground, and the jugs, get exactly 2 gallons of water into the 4-gallon jug.

Procedure :

1. Start with both jugs empty.
2. Fill the 3-gallon jug from the pump. It contains 3 gallons.
3. Pour all 3 gallons into the 4-gallon jug. The 4-gallon jug contains 3 gallons; the 3-gallon jug is empty.
4. Fill the 3-gallon jug again. It contains 3 gallons.
5. Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. This takes 1 gallon, leaving 2 gallons in the 3-gallon jug.
6. Empty the 4-gallon jug onto the ground.
7. Pour the 2 gallons remaining in the 3-gallon jug into the 4-gallon jug. The 4-gallon jug now contains exactly 2 gallons.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?
ii. Characterize the operating environment.
iii. What are the actions the agent can take?
iv. How can one evaluate the performance of the agent?
v. What sort of agent architecture do you think is most suitable for this agent? Why?

i. What can the rover sense?

1. It takes pictures of the land and rocks.
2. It senses obstacles and unsafe ground.
3. It checks what rocks and soil are made of.
4. It checks its location, direction, and power.

ii. What is the rover's environment like?

1. The rover works on Mars.
2. The ground is rocky, dusty, and uneven.
3. The rover cannot see everything around it at once.
4. It may face unknown dangers and cannot always contact Earth right away.

iii. What actions can the rover take?

1. Move and turn.
2. Stop or change direction to avoid danger.
3. Take pictures.
4. Study rocks and soil.
5. Send information to Earth.

iv. How can we judge its performance?

1. Did it travel safely?
2. Did it avoid obstacles?
3. Did it collect useful information and samples?
4. Did it save power and send its findings to Earth?

v. What kind of agent is suitable, and why?

A **goal-based agent** is suitable. It can make plans to reach places, study samples, and avoid danger. It can also work on its own when it cannot contact Earth.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Problem Statement

Place 8 queens on an 8 × 8 chessboard so that no two queens can attack each other. This means no two queens can be in the same row, column, or diagonal.

Procedure :

1. Draw an 8 × 8 chessboard.
2. Place one queen in the first column.
3. Place one queen in each of the remaining columns, making sure each queen is in a different row.
4. Check that no two queens are on the same diagonal.
5. If any queens share a diagonal, move one of them to a safe square in its column.
6. Continue checking and moving queens until no two queens can attack each other.
7. Confirm that the board has 8 queens and none share a row, column, or diagonal.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Problem-Solving Agent :

A **goal-based problem-solving agent** can be used. Its goal is to help the customer travel from the starting location to the destination using a suitable cab.

Pseudocode :

```
BEGIN
   Get the customer's starting location
   Get the customer's destination
   Get the customer's preferred cab type

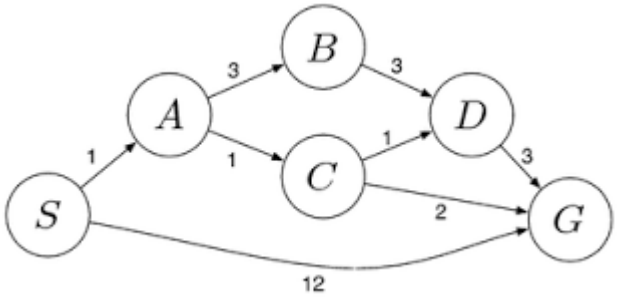
   Show available cabs and fares

   IF the preferred cab is available THEN
      Select the preferred cab
   ELSE
      Show other available cab types
      Ask the customer to select one
   END IF

   Confirm the booking
   Find a route to the destination
   Send the driver and cab details to the customer
END
```

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Problem Components :

1. **Initial state:** Starting warehouse **S**
2. **Goal state:** Destination warehouse **G**
3. **Actions:** Travel along a route from one warehouse to a connected warehouse.
4. **Path cost:** The sum of the route costs traveled.
5. **Goal:** Find a path from **S** to **G** with the lowest total cost.

Procedure :

1. Start at **S** with cost **0**.
2. From **S**, the available routes are:
   - **S → A**, cost **1**
   - **S → G**, cost **12**
3. Choose the route with the lowest total cost, **S → A** (cost **1**).
4. From **A**, the routes are:
   - **A → B**, giving total cost **1 + 3 = 4**
   - **A → C**, giving total cost **1 + 1 = 2**
5. Choose **A → C**, with total cost **2**.
6. From **C**, the routes are:
   - **C → D**, giving total cost **2 + 1 = 3**
   - **C → G**, giving total cost **2 + 2 = 4**
7. Choose **C → D**, with total cost **3**.
8. From **D**, the route **D → G** gives total cost **3 + 3 = 6**.
9. Compare the available goal routes: **S → G** costs **12**, **S → A → C → G** costs **4**, and **S → A → C → D → G** costs **6**.
10. Select the least-cost path: **S → A → C → G**, with total cost **4**.

Answer: The least-cost route is **S → A → C → G**, and its total cost is **4**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem